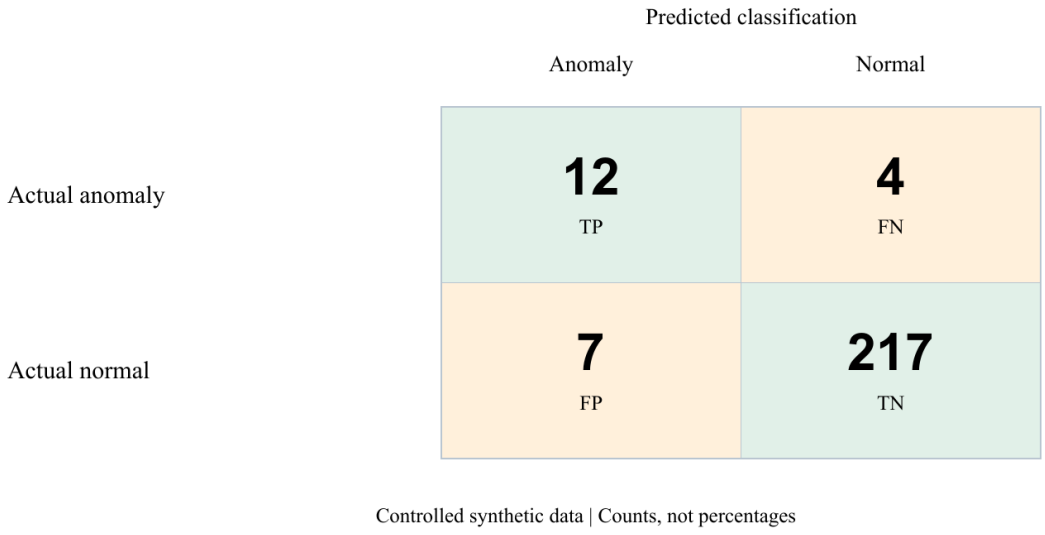
Controlled Synthetic Evaluation

Rainfall anomaly detection - synchronized local station cluster

Completed 27 October 2026. This report documents measured output from the unchanged application pipeline. Every station location and rainfall measurement in this evaluation is hypothetical. No independent historical-data accuracy is claimed.

Dataset property	Evaluated fixture
Station layout	8 fictional QC-area sites; two nearby quartets
Shared observation period	1-30 September 2023; all 24 hourly timestamps each day
Coverage	5,760 hourly readings -> 240 assessed station-days
Reference labels	8 high faults; 8 low faults; 224 valid readings
Distances	Selected neighbors 0.43-0.62 km; maximum pair distance 3.1513 km

Combined anomaly detection



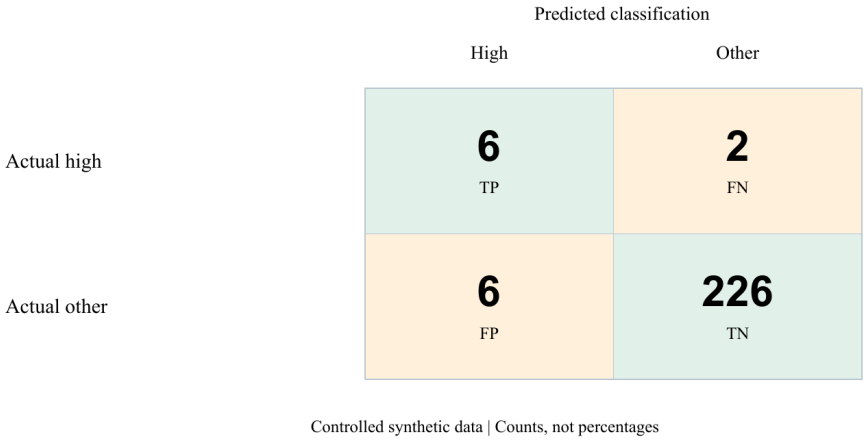
Controlled synthetic data | Counts, not percentages

Precision 63.16% | Recall 75.00% | F1 68.57% | Accuracy 95.42%

The detector found 12 of 16 injected faults, missed 4, and raised 7 false alarms among 224 valid readings. An always-normal predictor would already achieve 93.33% accuracy but 0% recall. Accuracy alone therefore overstates the usefulness of this detector.

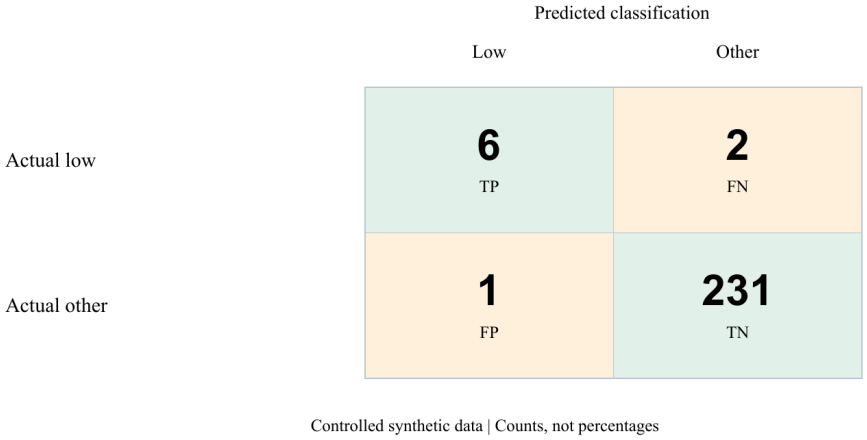
Detection direction and metrics

High-reading detection



Precision 50.00% | Recall 75.00% | F1 60.00% | Accuracy 96.67%

Low-reading detection



Precision 85.71% | Recall 75.00% | F1 80.00% | Accuracy 98.75%

Actual / predicted	Normal	High	Low
Normal	217	6	1
High	2	6	0
Low	2	0	6

High and low matrices are one-versus-rest. The high detector combines LOF with rainfall gates; the low detector uses separate rules. The combined result evaluates the full flagging system, rather than LOF alone. No flags in the wrong direction occurred in this run.

Repeatability and data quality

Fresh-process runs use the application's default seeded generator. Same-process runs retain its advancing jitter state. Changed-score counts use the scores exposed by the API schema. Wall times include local execution overhead and are not deployment benchmarks.

Run	Changed flags	Changed scores	Elapsed (s)
fresh_1	0	0	9.65
fresh_2	0	0	59.541
fresh_3	0	0	64.07
same_process_1	0	0	4.899
same_process_2	0	85	3.881
same_process_3	0	86	3.87

Input and assessment checks

Case	Status	Observed behavior
isolated missing hour	PASS	240 daily rows; 240 assessed; 0 unassessed
two missing hours	PASS	239 daily rows; 236 assessed; 3 unassessed
absent hourly timestamp	PASS	239 daily rows; 236 assessed; 3 unassessed
duplicate timestamp	PASS	240 daily rows; 240 assessed; 0 unassessed
negative rainfall	PASS	Invalid input rejected
infinite rainfall	PASS	Invalid input rejected
malformed timestamp	PASS	Invalid input rejected
three stations only	PASS	90 daily rows; 0 assessed; 90 unassessed
nonoverlapping station dates	PASS	240 daily rows; 120 assessed; 120 unassessed
all dry control	PASS	240 daily rows; 240 assessed; 0 unassessed
uniform heavy rain control	PASS	240 daily rows; 240 assessed; 0 unassessed

These quality variants are separate engineering checks; they are not pooled into the 240-reading confusion matrices. An unassessed reading is never treated as a confirmed normal prediction in the detection evaluation.

Route tests and verification

Extended check groups: 22 passed, 0 failed. Passing these checks validates the evaluation and documented system behavior; it does not erase the detection errors.

HTTP check	Observed status
synchronous multipart upload	200
async upload and owner polling	202 -> done
non-CSV	415
empty CSV	400
oversized content length	413
malformed CSV	422
missing files	422

Existing backend regression suite

Status: PASS. Existing tests executed: 54. The complete test log is backend_tests.txt. It covers existing quality, station review, history, event-stream and search checks.

Verification scope

Synchronous multipart upload and async job polling exercised the actual local FastAPI router and pipeline. Authentication was supplied through a synthetic dependency override and rate limiting was disabled only in the local test. No actual user login, browser UI, production deployment, mobile device or database persistence was tested. The HTTP checks are not an authentication or security certification.

Evidence chain

The original manifest records source/input/prediction hashes, package versions and detector thresholds. The extended package retains repeat outputs, all paired classifications, per-station and per-scenario breakdowns, quality-case observations, route results and regression logs. Checksums establish which files were evaluated; they are not external certification of the reference labels.

Limits on interpretation

Reference labels identify deliberately injected sensor interventions and valid weather by construction. The scenarios were designed with knowledge of the existing detector, so this is a development functional evaluation, not a blinded holdout. The six-hour rainfall profiles are not calibrated meteorological simulations. There are only 16 fault observations, including correlated station-days; the percentages do not establish real AWS performance. The manuscript's historical-data evaluation still requires independently verified historical labels.

Misclassifications and discussion

Date	Station	Scenario	Actual -> predicted
2023-09-09	QCA01	valid localized shower	normal -> high
2023-09-15	QCB03	zero fault on light rain	low -> normal
2023-09-16	QCB04	progressive overreport	high -> normal
2023-09-17	QCA02	shared rain with variation	normal -> high
2023-09-18	QCB02	valid localized shower	normal -> high
2023-09-18	QCB04	progressive overreport	high -> normal
2023-09-19	QCB03	shared rain with variation	normal -> high
2023-09-20	QCA03	shared rain with variation	normal -> high
2023-09-20	QCB02	partial underreport	low -> normal
2023-09-21	QCA04	valid localized dry patch	normal -> low
2023-09-29	QCA01	shared rain with variation	normal -> high

Suggested Results and Discussion text

The controlled evaluation comprised 240 station-day observations from eight hypothetical nearby stations with synchronized hourly records. Reference labels specified 16 injected fault readings and 224 valid readings. The system correctly detected 12 faults, missed four, and incorrectly flagged seven valid readings. Combined precision, recall, F1-score and accuracy were 63.16%, 75.00%, 68.57% and 95.42%, respectively. High-reading precision was 50.00%, compared with 85.71% for low-reading detection; both achieved 75.00% recall. These results apply only to the declared synthetic scenarios.

False alarms occurred on valid localized showers, a valid dry patch and ordinary station variation. Missed faults included progressive overreporting, a zero fault during light rainfall and partial underreporting above the low rule's 1 mm ceiling. The findings show that nearby stations and complete temporal overlap enable comparisons, but spatial disagreement alone does not distinguish sensor error from legitimate local weather. The observed errors should be retained and reported rather than removed by relabeling test cases.

Reproduction

Run run_extended_testing.py using the API virtual environment. Run render_report.py using the bundled ReportLab runtime. Both scripts and their required evidence files are included. The original generator remains separate so reference data can be preserved without regeneration. The application algorithms and thresholds were not modified.